

Python List, Tuple, Dictionary & Set Programs

Programs and outputs compiled from the supplied notes.

1. Multiple Datatype List

Program:

```
n_list = [2.9, 3000, "Hello", "Welcome to python list", True]
print(n_list)
```

Output:

```
[2.9, 3000, 'Hello', 'Welcome to python list', True]
```

2. Nested List

Program:

```
n_list = ["Happy", [2, 0, 1, 5]]
print(n_list[0][3])
print(n_list[1][2])
```

Output:

```
p
1
```

3. Nested List Example

Program:

```
my_list = ["mouse", [8, 4, 6], ['a', 'b', 'c']]
print(my_list)
print(my_list[0])
print(my_list[1][1])
print(my_list[2][2])
```

Output:

```
['mouse', [8, 4, 6], ['a', 'b', 'c']]
mouse
4
c
```

4. Negative Indexing

Program:

```
my_list = ['p', 'r', 'o', 'b', 'e']
print(my_list[-1])
print(my_list[-5])
```

Output:

```
e
p
```

5. Append and Extend

Program:

```
odd = [1, 3, 5]
odd.append(7)
print(odd)
odd.extend([9, 11, 13])
print(odd)
```

Output:

```
[1, 3, 5, 7]
[1, 3, 5, 7, 9, 11, 13]
```

6. Concatenating and Repeating

Program:

```
odd = [1, 3, 5]
print(odd + [9, 7, 5])
print(["re"] * 3)
```

Output:

```
[1, 3, 5, 9, 7, 5]
['re', 're', 're']
```

7. List Length and Remove

Program:

```
thislist = ["apple", "banana", "cherry"]
```

```
print(len(thislist))
thislist = ["apple", "banana", "cherry"]
thislist.remove("banana")
print(thislist)
```

Output:

```
3
['apple', 'cherry']
```

8. Sort List

Program:

```
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]
thislist.sort()
print(thislist)
thislist = [100, 50, 65, 82, 23]
thislist.sort()
print(thislist)
```

Output:

```
['banana', 'kiwi', 'mango', 'orange', 'pineapple']
[23, 50, 65, 82, 100]
```

9. Sort Descending

Program:

```
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]
thislist.sort(reverse=True)
print(thislist)
thislist = [100, 50, 65, 82, 23]
thislist.sort(reverse=True)
print(thislist)
```

Output:

```
['pineapple', 'orange', 'mango', 'kiwi', 'banana']
[100, 82, 65, 50, 23]
```

10. Insert

Program:

```
fruits = ["apple", "banana", "cherry"]
fruits.insert(1, "orange")
print(fruits)
```

Output:

```
['apple', 'orange', 'banana', 'cherry']
```

11. Pop

Program:

```
thislist = ["apple", "banana", "cherry"]
thislist.pop(1)
print(thislist)
```

Output:

```
['apple', 'cherry']
```

12. Del Operator

Program:

```
thislist = ["apple", "banana", "cherry"]
del thislist[0]
print(thislist)
```

Output:

```
['banana', 'cherry']
```

13. Clear List

Program:

```
thislist = ["apple", "banana", "cherry"]
thislist.clear()
print(thislist)
```

Output:

```
[]
```

14. Count Method

Program:

```
fruits = ["apple", "banana", "cherry", "apple"]
```

```
x = fruits.count("apple")
print(x)
fruits = [1, 4, 2, 9, 7, 8, 9, 3, 1]
x = fruits.count(9)
print(x)
```

Output:

```
2
2
```

15. Copy Method

Program:

```
lst = [10, 20, 30, 40]
new_lst = lst.copy()
print("Original List :", lst)
print("Copied List :", new_lst)
```

Output:

```
Original List : [10, 20, 30, 40]
Copied List : [10, 20, 30, 40]
```

16. Index Method

Program:

```
lst = [10, 20, 30, 40, 50]
value = 30
position = lst.index(value)
print("List :", lst)
print("Value :", value)
print("Index :", position)
```

Output:

```
List : [10, 20, 30, 40, 50]
Value : 30
Index : 2
```

17. List Built-in Functions

Program:

```
nums = [3, 41, 12, 9, 74, 15]
print(len(nums))
print(max(nums))
print(min(nums))
print(sum(nums))
print(sum(nums) / len(nums))
```

Output:

```
6
74
3
154
25.666666666666668
```

18. String to List

Program:

```
s = "spam"
t = list(s)
print(t)
```

Output:

```
['s', 'p', 'a', 'm']
```

19. Split

Program:

```
s = "pining for the fjords"
t = s.split()
print(t)
print(t[2])
```

Output:

```
['pining', 'for', 'the', 'fjords']
the
```

20. Split Using Hyphen

Program:

```
s = "spam-spam-spam"
delimiter = "-"
```

```
print(s.split(delimiter))
```

Output:

```
['spam', 'spam', 'spam']
```

21. Tuple

Program:

```
my_data = ("hi", "hello", "bye")
print(my_data)
my_data2 = (1, 2.8, "Hello World")
print(my_data2)
```

Output:

```
('hi', 'hello', 'bye')
(1, 2.8, 'Hello World')
```

22. Accessing Tuple Elements

Program:

```
my_data = ("hi", "hello", "bye")
print(my_data)
print(my_data[0])
print(my_data[2])
```

Output:

```
('hi', 'hello', 'bye')
hi
bye
```

23. Negative Indexing in Tuple

Program:

```
my_data = (1, 2, "Kevin", 8.9)
print(my_data[-1])
print(my_data[-3])
```

Output:

```
8.9
2
```

24. Nested Tuples

Program:

```
my_data = (1, "Steve", (11, 22, 33))
print(my_data[1][3])
print(my_data[2][1])
```

Output:

```
v
22
```

25. Changing Elements of Tuple

Program:

```
my_data = (1, [9, 8, 7], "World")
print(my_data)
my_data[1][2] = 99
print(my_data)
my_data[0] = 101
print(my_data)
```

Output:

```
(1, [9, 8, 7], 'World')
(1, [9, 8, 99], 'World')
TypeError: 'tuple' object does not support item assignment
```

26. Delete Operation on Tuple

Program:

```
my_data = (1, 2, 3, 4, 5, 6)
print(my_data)
del my_data
```

Output:

```
(1, 2, 3, 4, 5, 6)
```

27. Slicing Operation in Tuples

Program:

```
my_data = (11, 22, 33, 44, 55, 66, 77, 88, 99)
print(my_data)
print(my_data[2:5])
print(my_data[:4])
print(my_data[4:])
print(my_data[4:-1])
print(my_data[:])
```

Output:

```
(11, 22, 33, 44, 55, 66, 77, 88, 99)
(33, 44, 55)
(11, 22, 33, 44)
(55, 66, 77, 88, 99)
(55, 66, 77, 88)
(11, 22, 33, 44, 55, 66, 77, 88, 99)
```

28. Membership Test in Tuples

Program:

```
my_data = (11, 22, 33, 44, 55, 66, 77, 88, 99)
print(my_data)
print(22 in my_data)
print(2 in my_data)
print(88 not in my_data)
```

Output:

```
(11, 22, 33, 44, 55, 66, 77, 88, 99)
True
False
False
```

29. Iterating a Tuple

Program:

```
my_tuple = ("Apple", "Orange", "Grapes", "Banana")
for fruit in my_tuple:
    print(fruit)
```

Output:

```
Apple
Orange
Grapes
Banana
```

30. Basic Tuple Operations

Program:

```
print(len((1, 2, 3)))
print((1, 2, 3) + (4, 5, 6))
print(("Hi!",) * 4)
print(3 in (1, 2, 3))
for x in (1, 2, 3):
    print(x)
```

Output:

```
3
(1, 2, 3, 4, 5, 6)
('Hi!', 'Hi!', 'Hi!', 'Hi!')
True
1
2
3
```

31. Tuple len() Method

Program:

```
tuple1, tuple2 = (123, 'xyz', 'zara'), (456, 'abc')
print("First tuple length : ", len(tuple1))
print("Second tuple length : ", len(tuple2))
```

Output:

```
First tuple length : 3
Second tuple length : 2
```

32. Tuple max() Method

Program:

```
tuple1, tuple2 = ('xyz', 'zara', 'abc'), (456, 700, 200)
print("Max value element : ", max(tuple1))
print("Max value element : ", max(tuple2))
```

Output:

```
Max value element : zara
Max value element : 700
```

33. Tuple Packing and Unpacking

Program:

```
b = ("Bob", 19, "CS")
(name, age, studies) = b
print(name)
print(age)
print(studies)
```

Output:

```
Bob
19
CS
```

34. Accessing Dictionary Elements

Program:

```
my_dict = {'name': 'Jack', 'age': 26}
print(my_dict['name'])
print(my_dict.get('age'))
```

Output:

```
Jack
26
```

35. Creating Dictionaries

Program:

```
my_dict = {}
print("1. Empty Dictionary:", my_dict)
my_dict = {1: 'apple', 2: 'ball'}
print("2. Integer Keys:", my_dict)
my_dict = {'name': 'John', 1: [2, 4, 3]}
print("3. Mixed Keys:", my_dict)
my_dict = dict({1: 'apple', 2: 'ball'})
print("4. Using dict():", my_dict)
my_dict = dict([(1, 'apple'), (2, 'ball')])
print("5. From Sequence:", my_dict)
```

Output:

```
1. Empty Dictionary: {}
2. Integer Keys: {1: 'apple', 2: 'ball'}
3. Mixed Keys: {'name': 'John', 1: [2, 4, 3]}
4. Using dict(): {1: 'apple', 2: 'ball'}
5. From Sequence: {1: 'apple', 2: 'ball'}
```

36. Changing and Adding Dictionary Elements

Program:

```
my_dict = {'name': 'Jack', 'age': 26}
my_dict['age'] = 27
print(my_dict)
my_dict['address'] = 'Downtown'
print(my_dict)
```

Output:

```
{'name': 'Jack', 'age': 27}
{'name': 'Jack', 'age': 27, 'address': 'Downtown'}
```

37. Dictionary Example

Program:

```
tinydict = {'name': 'john', 'code': 6734, 'dept': 'sales'}
print(tinydict)
print(tinydict.keys())
print(tinydict.values())
```

Output:

```
{'name': 'john', 'code': 6734, 'dept': 'sales'}
dict_keys(['name', 'code', 'dept'])
dict_values(['john', 6734, 'sales'])
```

38. Removing Dictionary Elements

Program:

```
squares = {1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
print(squares.pop(4))
print(squares)
squares.clear()
print(squares)
del squares
print(squares)
```

Output:

```
16
{1: 1, 2: 4, 3: 9, 5: 25}
{}
NameError: name 'squares' is not defined
```

39. Membership Test for Dictionary Keys

Program:

```
squares = {1: 1, 3: 9, 5: 25, 7: 49, 9: 81}
print(1 in squares)
print(2 not in squares)
```

```
print(49 in squares)
```

Output:

True

True

False

40. Set of Integers

Program:

```
my_set = {1, 2, 3}
print(my_set)
```

Output:

```
{1, 2, 3}
```

41. Set of Mixed Datatypes

Program:

```
my_set = {1.0, "Hello", (1, 2, 3)}
print(my_set)
```

Output:

```
{1.0, 'Hello', (1, 2, 3)}
```

42. Set Does Not Have Duplicates

Program:

```
my_set = {1, 2, 3, 4, 3, 2}
print(my_set)
```

Output:

```
{1, 2, 3, 4}
```

43. Adding Element

Program:

```
my_set = {1, 2, 3}
my_set.add(4)
print(my_set)
```

Output:

```
{1, 2, 3, 4}
```

44. Removing Element

Program:

```
my_set = {1, 2, 3}
my_set.remove(2)
print(my_set)
```

Output:

```
{1, 3}
```

45. From List to Set

Program:

```
numbers = [1, 2, 2, 3, 4]
my_set = set(numbers)
print(my_set)
```

Output:

```
{1, 2, 3, 4}
```

46. Set Union

Program:

```
nums1 = {1, 2, 2, 3, 4, 5}
nums2 = {4, 5, 6, 7, 7, 8}
distinct_nums = nums1.union(nums2)
print("The union of two sets is:", distinct_nums)
```

Output:

```
The union of two sets is: {1, 2, 3, 4, 5, 6, 7, 8}
```

47. Set Intersection

Program:

```
nums1 = {1, 2, 2, 3, 4, 5}
nums2 = {4, 5, 6, 7, 7, 8}
common_nums = nums1.intersection(nums2)
print("The intersection of two sets is:", common_nums)
```

Output:

```
The intersection of two sets is: {4, 5}
```

48. Set Difference

Program:

```
nums1 = {1, 2, 2, 3, 4, 5}
nums2 = {4, 5, 6, 7, 7, 8}
nums3 = nums1.difference(nums2)
nums4 = nums2.difference(nums1)
print("nums1 - nums2:", nums3)
print("nums2 - nums1:", nums4)
```

Output:

```
nums1 - nums2: {1, 2, 3}
nums2 - nums1: {8, 6, 7}
```

49. Set Operations

Program:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
print(A | B)
print(A & B)
print(A - B)
```

Output:

```
{1, 2, 3, 4, 5, 6}
{3, 4}
{1, 2}
```